

Documentation: Automated Malicious IP Detection and Email Alerting

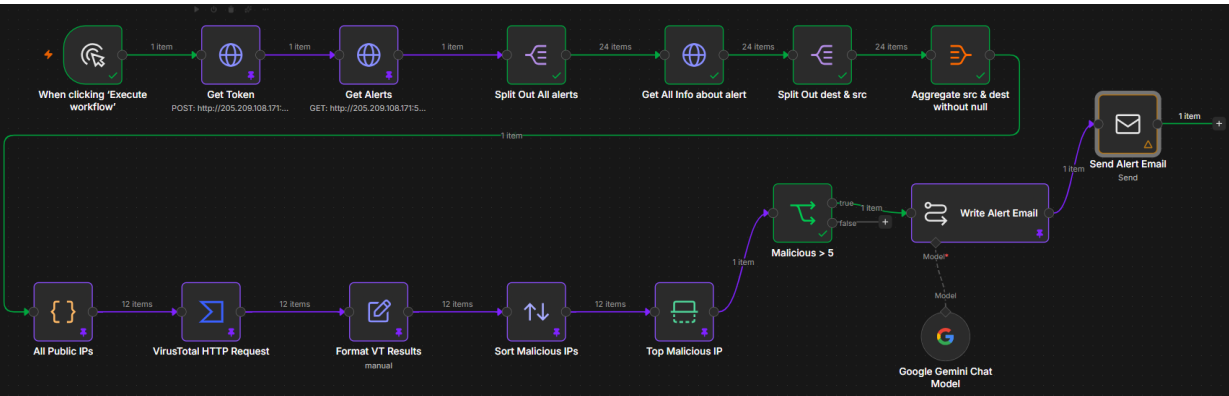
1. Overview

Goal: Automate the process of scanning the IPs found in security alerts, identify the most dangerous malicious IP using VirusTotal, generate a professional alert email with AI, and send it automatically.

Tools used:

Tool	Purpose
n8n	Building and running the workflow
Alerts system API	Fetching alerts and their details
VirusTotal API v3	IP reputation lookup
Google Gemini	Writing the email body in HTML
SMTP (Gmail)	Sending the email

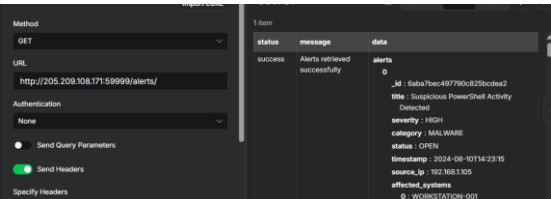
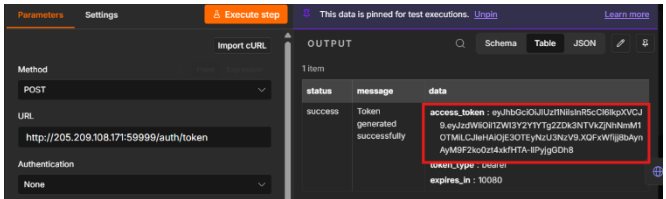
2. Workflow Diagram



3. Node-by-Node Explanation

3.1 Fetching the Alerts

- **Get Token:** POST request to obtain the authentication token.
- **Get Alerts:** GET request to retrieve the list of alerts using the token.



3.2 Extracting Alert Details

- **Split Out All alerts:** Splits each alert into its own item (24 items).

Parameters Settings **Execute step** OUTPUT 24 items

Fields To Split Out

data.alerts

Use \$binary to split out the input item by binary data

Include

All Other Fields

Options

total_count : 24

data.alerts

_id : 6aba7bec497790c825bcdca2

title : Suspicious PowerShell Activity Detected

severity : HIGH

category : MALWARE

status : OPEN

- **Get All Info about alert:** Fetches the full details of each alert.

Import cURL

Method

GET

URL

http://205.209.108.171:59999/alerts/{{ \$json['data.alerts']._id }}

Authentication

None

24 items

status message data

success Alert retrieved successfully

_id : 6aba7bec497790c825bcdca2

title : Suspicious PowerShell Activity Detected

severity : HIGH

category : MALWARE

status : OPEN

timestamp : 2024-08-10T14:23:15

source_ip : 192.168.1.105

source_port : null

destination_ip : 203.0.113.45

destination_port : 443

- **Split Out dest & src:** Separates the source_ip and destination_ip fields.

Parameters Settings **Execute step** OUTPUT 24 items

Fields To Split Out

data.source_ip, data.destination_ip

Use \$binary to split out the input item by binary data

Include

No Other Fields

Options

+ Add Field

data.source_ip

data.destination_ip

192.168.1.105

203.0.113.45

null

null

192.168.1.203

142.250.191.78

null

null

94.102.49.190

10.0.3.25

null

167.172.203.44

- **Aggregate src & dest without null:** Collects all IPs in one place and ignores empty values.

Parameters Settings **Execute step** OUTPUT 1 item

Aggregate

Individual Fields

Fields To Aggregate

data.source_ip

data.destination_ip

Input Field Name

data.source_ip

Enter the field name as text

Rename Field

Input Field Name

data.destination_ip

Enter the field name as text

source_ip

destination_ip

0 : 192.168.1.105

0 : 203.0.113.45

1 : 192.168.1.203

1 : 142.250.191.78

2 : 94.102.49.190

2 : 10.0.3.25

3 : 192.168.1.77

3 : 167.172.203.44

4 : 203.0.113.156

4 : 104.21.35.147

5 : 45.142.214.89

5 : 10.0.1.45

6 : 192.168.2.87

6 : 10.0.1.50

7 : 172.16.0.199

7 : 198.51.100.123

8 : 203.0.113.78

8 : 172.16.0.0/24

9 : 192.168.3.44

9 : 10.0.0.5

10 : 192.168.1.155

10 : 8.8.8.8

11 : 192.168.1.45

11 : 10.0.2.25

12 : 192.168.2.134

12 : 192.168.1.78

13 : 103.224.182.251

13 : 195.220.101.42

14 : pool.minergate.com

15 : 10.0.3.15

3.3 Filtering Public IPs (Code node)

Function: Merges the IPs, removes duplicates, excludes private ranges (10.x, 172.16-31.x, 192.168.x, 127.x), and outputs **one item per IP**.

The screenshot shows a workflow editor with a JavaScript code node. The code is designed to process an array of items, each containing a source IP and a destination IP. It merges these into a single array, removes duplicates, and filters out private IP ranges. The output is a list of 12 unique public IP addresses.

```
1 const ips = [];  
2  
3 for (const item of $input.all()) {  
4   const source = item.json.source_ip;  
5   const destination = item.json.destination_ip;  
6  
7   if (source) {  
8     if (Array.isArray(source)) {  
9       ips.push(...source);  
10    } else {  
11      ips.push(source);  
12    }  
13  }  
14 }
```

OUTPUT

ip
94.102.49.190
203.0.113.156
45.142.214.89
203.0.113.78
103.224.182.251
203.0.113.45
142.250.191.78
167.172.203.44
104.21.35.147
198.51.100.123
8.8.8.8
185.220.101.42

3.4 VirusTotal Lookup

The screenshot shows a workflow editor with a HTTP request node. The URL is configured to use the IP address from the previous step to look up information on VirusTotal. The output shows the details for the IP 94.102.49.190, including its type, links, and attributes.

Use the [VirusTotal docs](#) to construct your request. We'll take care of the authentication part if you add a VirusTotal credential below.

Method: GET

URL: `https://www.virustotal.com/api/v3/ip_addresses/{{ $json.ip }}`

Output:

```
{  
  "data": {  
    "id": "94.102.49.190",  
    "type": "ip_address",  
    "links": {  
      "self": "https://www.virustotal.com/api/v3/ip_addresses/94.102.49.190"  
    },  
    "attributes": {  
      "regional_internet_registry": "RIPE NCC",  
      "reputation": 6,  
      "continent": "EU"  
    }  
  }  
}
```

3.5 Formatting the Results (Edit Fields)

The screenshot shows a workflow editor with an Edit Fields node. The node is configured to map the output of the previous step into a table with columns for IP address, malicious status, suspicious status, harmless status, undetected status, country, owner, and reputation. The output is a table with 12 rows of data.

Mode: Manual Mapping

Fields to Set:

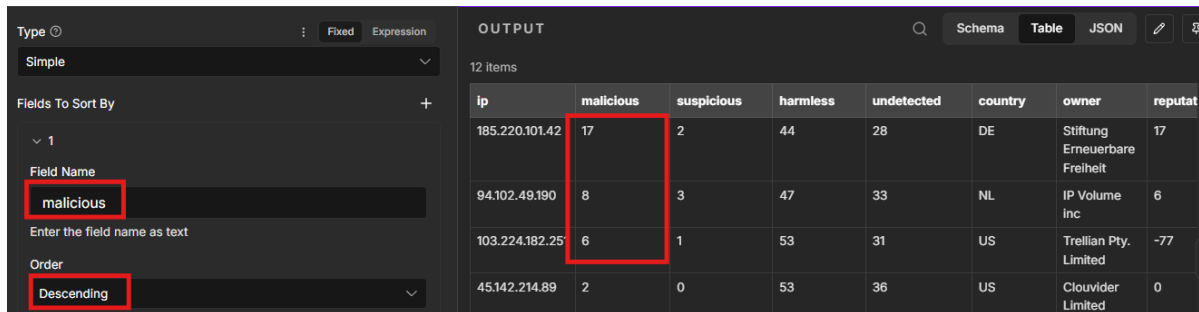
- ip: `{{ $json.data.id }}`
- malicious: `{{ $json.data.attributes.last_analysis_stats.malicious }}`
- suspicious: `{{ $json.data.attributes.last_analysis_stats.suspicious }}`

OUTPUT

ip	malicious	suspicious	harmless	undetected	country	owner	reputat
94.102.49.190	8	3	47	33	NL	IP Volume Inc	6
203.0.113.156	0	0	54	37	null	null	0
45.142.214.89	2	0	53	36	US	Clouvider Limited	0
203.0.113.78	0	0	54	37	null	null	0
103.224.182.251	6	1	53	31	US	Trellian Pty. Limited	-77
203.0.113.45	0	0	54	37	null	null	2
142.250.191.78	0	0	55	36	US	Google LLC	0
167.172.203.44	0	0	54	37	US	DigitalOcean, LLC	0
104.21.35.147	0	0	55	36	null	Cloudflare, Inc.	0
198.51.100.123	0	0	54	37	null	null	0
8.8.8.8	0	0	53	38	US	Google LLC	561
185.220.101.42	17	2	44	28	DE	Stiftung Erneuerbare	17

3.6 Selecting the Most Dangerous IP

Selection criterion: The IP with the **highest malicious count**. This value is the number of security engines on VirusTotal (out of roughly 90) that flagged the IP as malicious. The higher the number, the stronger the agreement between engines that the IP is dangerous.

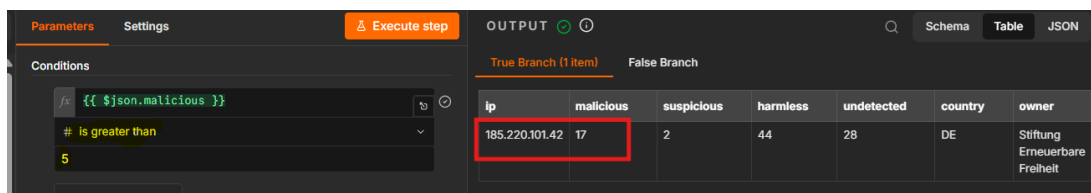


The screenshot shows a workflow node configuration on the left and an output table on the right. In the configuration, 'malicious' is selected as the field to sort by, and the order is set to 'Descending'. The output table displays 12 items with columns: ip, malicious, suspicious, harmless, undetected, country, owner, and reputat. The first row, with IP 185.220.101.42 and a malicious count of 17, is highlighted with a red box.

ip	malicious	suspicious	harmless	undetected	country	owner	reputat
185.220.101.42	17	2	44	28	DE	Stiftung Erneuerbare Freiheit	17
94.102.49.190	8	3	47	33	NL	IP Volume Inc	6
103.224.182.25	6	1	53	31	US	Trellian Pty. Limited	-77
45.142.214.89	2	0	53	36	US	Clouvider Limited	0

3.7 Alert Condition (IF)

- Condition: `{{ $json.malicious }}` **is greater than** 5
- The **true** output continues to the AI node; the **false** output is left unconnected (the workflow stops without sending anything).
- The value 5 is the **threshold** and can be adjusted based on the desired sensitivity.

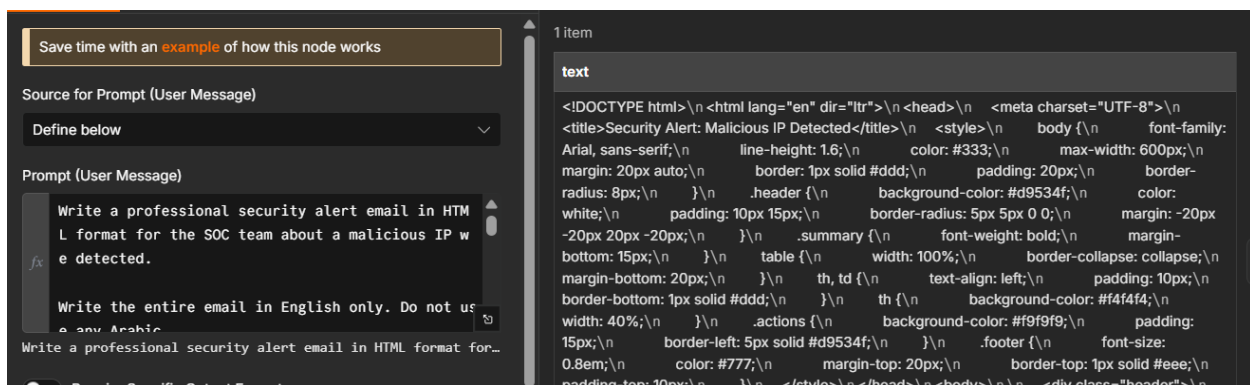


The screenshot shows a workflow node configuration on the left and an output table on the right. In the configuration, the condition is set to 'is greater than' with a value of 5. The output table displays the 'True Branch' with 1 item, where the first row (IP 185.220.101.42, malicious count 17) is highlighted with a red box.

ip	malicious	suspicious	harmless	undetected	country	owner
185.220.101.42	17	2	44	28	DE	Stiftung Erneuerbare Freiheit

3.8 Writing the Email with AI

- **Type:** Basic LLM Chain + Google Gemini Chat Model (gemini-2.5-flash)
- **Settings:** Source for Prompt = Define below (Expression)



The screenshot shows a workflow node configuration on the left and an output table on the right. In the configuration, the 'Source for Prompt' is set to 'Define below' and the 'Prompt' is a multi-line instruction to write a professional security alert email in HTML format. The output table displays 1 item with a 'text' field containing a detailed HTML email template.

Source for Prompt (User Message)

Define below

Prompt (User Message)

Write a professional security alert email in HTML format for the SOC team about a malicious IP detected.

Write the entire email in English only. Do not use any Arabic.

Write a professional security alert email in HTML format for...

Require Specific Output Format

text
<!DOCTYPE html>\n<html lang="en" dir="ltr">\n<head>\n<meta charset="UTF-8">\n<title>Security Alert: Malicious IP Detected</title>\n<style>\nbody {\nfont-family: Arial, sans-serif;\nline-height: 1.6;\ncolor: #333;\nmax-width: 600px;\nmargin: 20px auto;\nborder: 1px solid #ddd;\npadding: 20px;\nborder-radius: 8px;\n}\n.header {\nbackground-color: #d9534f;\ncolor: white;\npadding: 10px 15px;\nborder-radius: 5px 5px 0 0;\nmargin: -20px -20px 20px -20px;\n}\n.summary {\nfont-weight: bold;\nmargin-bottom: 15px;\n}\n.table {\nwidth: 100%;\nborder-collapse: collapse;\nmargin-bottom: 20px;\n}\nth, td {\ntext-align: left;\npadding: 10px;\nborder-bottom: 1px solid #ddd;\n}\nth {\nbackground-color: #f4f4f4;\nwidth: 40%;\n}\n.actions {\nbackground-color: #f9f9f9;\npadding: 15px;\nborder-left: 5px solid #d9534f;\n}\n.footer {\nfont-size: 0.8em;\ncolor: #777;\nmargin-top: 20px;\nborder-top: 1px solid #eee;\npadding-top: 10px;\n}\n</style>\n</head>\n<body>\n\n<div class="header">\n

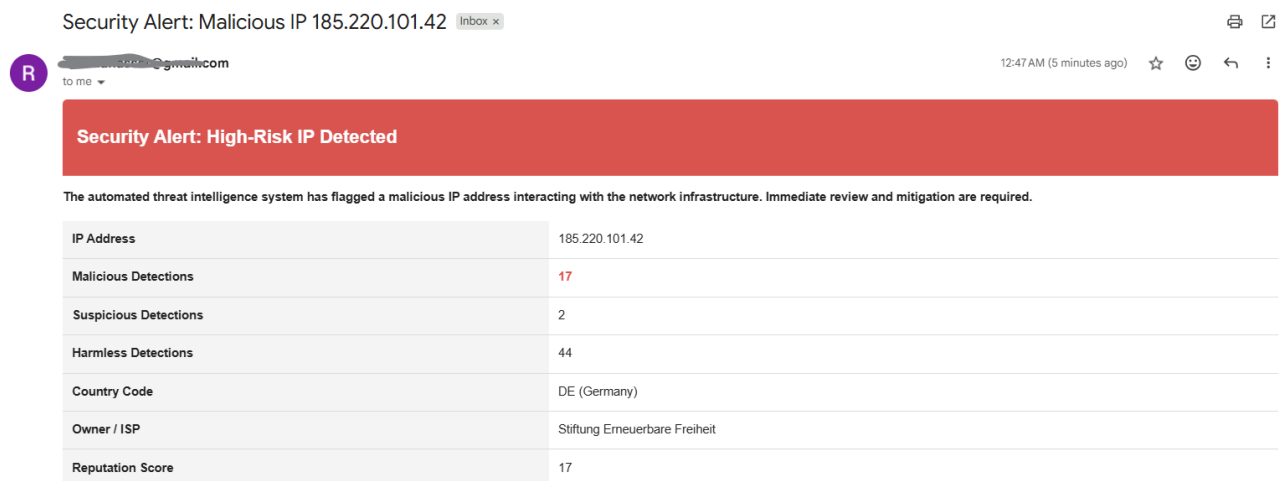
3.9 Sending the Email

Field	Value
Credential	SMTP (smtp.gmail.com, Port 465, SSL/TLS ON)
From Email	Sender address
To Email	Recipient address
Subject	Security Alert: Malicious IP {{ \$("Top Malicious IP").item.json.ip }}
Email Format	HTML
HTML	{{ \$json.text }}

4. Final Result

The email arrived with the subject `Security Alert: Malicious IP 185.220.101.42` and contains:

- A red header with the alert title
- A table with the IP details (detection count, country, owner, reputation)
- Recommended actions (block the IP and review logs)



Alert email received in the inbox

The workflow generated and delivered a professional HTML security alert for the most dangerous IP found, `185.220.101.42`. The email includes:

- **Header:** A red banner with the alert title ("Security Alert: Malicious IP Detected"), so the severity is clear at a glance.
- **Introduction:** A short summary addressed to the SOC team, explaining that multiple security engines flagged the IP as malicious.
- **Details table:** The key VirusTotal data for the IP:

Complete n8n workflow

The workflow runs end to end in a single execution:

1. **Collect:** Fetches the alerts and extracts their source and destination IPs.
2. **Enrich:** Filters the public IPs (12), checks them on VirusTotal, and keeps the one with the most malicious detections.
3. **Alert:** If that IP has more than 5 detections, Gemini writes an HTML alert email and it is sent to the SOC team. Otherwise, the workflow stops without sending anything.

Outcome: The most dangerous IP found was 185.220.101.42 (17 malicious detections), and the alert email was generated and delivered automatically.

